1. What happens if we stack multiple linear layers without activation?

   Answer: If we stack linear layers without an activation function like ReLU or Sigmoid between them, they act as a single linear layer. Without activation functions to "bend" the data, our deep, expensive network can only solve the same simple problems as a basic linear model.

2. What kind of decision boundary does an MLP create?

   Answer: A Multilayer Perceptron (MLP) with activation functions creates a non-linear decision boundary. A simple linear model can only draw a straight line to separate classes. An MLP can draw curves, circles, or complex zig-zags to wrap around data points that aren't neatly separated by a single straight line.

3. What are the main components of a neural network?

   Answer: There are three "physical" parts and one "process" part:

   Neurons (Nodes): The little units that hold numbers.

   Weights and Biases: The "knobs" the network turns to learn. Weights determine how much importance to give an input, and biases help shift the result.

   Layers: Groups of neurons (Input, Hidden, and Output).

   Activation Functions: The logic gates that decide if a neuron should (fire) pass information or stay quiet.

4. What is the difference between input layer, hidden layer, and output layer?

   Answer: The differences of input, hidden and output layer:

   Input Layer: These are the raw materials (e.g. the pixels of a photo). It doesn't do any math, it just receives the data.

   Hidden Layer(s): This is where the work happens. These layers sit in the middle and extract features (e.g. one layer looks for edges, the next for shapes, the next for eyes). You don't see this happening directly, which is why they are called "hidden."

   Output Layer: This is the finished product or the final decision (e.g. "This photo is 90% likely to be a cat").

5. What is forward propagation?

   Answer: Forward propagation is the process of data moving through the network in one direction from the input to the output. During forward propagation, the network is just "guessing" based on what it currently knows.